

Statistical precision of EHands Sum operator.

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Problem 1

Let

$$\hat{p} = \frac{X}{N}, \quad X \sim \text{Binomial}(N, p). \quad (1)$$

Define

$$y = 1 - 2p, \quad \hat{y} = 1 - 2\hat{p}. \quad (2)$$

How many shots N are needed to determine the sign of y with k -sigma confidence?

Solution 1

Under binomial sampling,

$$\sigma(\hat{p}) = \sqrt{\frac{p(1-p)}{N}}, \quad \sigma(\hat{y}) = 2\sqrt{\frac{p(1-p)}{N}}. \quad (3)$$

Require

$$|y| > k \sigma(\hat{y}). \quad (4)$$

Using $y = 1 - 2p$,

$$\boxed{N > \frac{4k^2 p(1-p)}{(1-2p)^2}} \quad (5)$$

Problem 2

Let

$$y = \frac{x-t}{2}, \quad (6)$$

with $x, t \in [-1, 1]$, and assume

$$y = 1 - 2p. \quad (7)$$

How many shots N are needed to determine the sign of y with k -sigma precision as a function of x and t ?

Solution 2

Using $p = (1 - y)/2$ in Solution 1,

$$N > k^2 \frac{1 - y^2}{y^2}. \quad (8)$$

With $y = (x - t)/2$,

$$\boxed{N > k^2 \frac{4 - (x - t)^2}{(x - t)^2}} \quad (9)$$

for $x \neq t$.

When $x = t$, the sign of y cannot be resolved with any finite number of shots.

Problem 3

Let

$$x' = \frac{x + 1}{2}, \quad t' = \frac{t + 1}{2}, \quad (10)$$

so $x' \in [0, 1]$ and restrict $t' \in [0, 1]$, equivalently $t \in [-1, 1]$. Consider an alternative circuit,

$$y' = wx' + (1 - w), \quad (11)$$

measured as

$$y' = 1 - 2p'. \quad (12)$$

Find $w = w(t')$ such that

$$y > 0 \iff y' > \frac{1}{2}, \quad (13)$$

and then compute the shots needed to decide whether $y' > 1/2$ with k -sigma certainty.

Solution 3

Since $y = (x - t)/2$, we have

$$y > 0 \iff x > t \iff x' > t'. \quad (14)$$

To make $y' > 1/2$ equivalent to $x' > t'$, place the threshold at $x' = t'$:

$$wt' + (1 - w) = \frac{1}{2}. \quad (15)$$

Hence

$$\boxed{w(t') = \frac{1}{2(1 - t')} = \frac{1}{1 - t}} \quad (16)$$

and

$$y' - \frac{1}{2} = \frac{x' - t'}{2(1 - t')}. \quad (17)$$

Therefore

$$y' > \frac{1}{2} \iff x' > t' \iff y > 0. \quad (18)$$

For the statistical precision, require

$$y' - \frac{1}{2} > k \sigma(\hat{y}'), \quad (19)$$

with

$$\sigma(\hat{y}') = 2\sqrt{\frac{p'(1-p')}{N'}} = \sqrt{\frac{1-(y')^2}{N'}}. \quad (20)$$

This gives

$$N' > k^2 \frac{1-(y')^2}{\left(y' - \frac{1}{2}\right)^2}. \quad (21)$$

Substituting $y' = wx' + (1-w)$ and $w = 1/[2(1-t')]$,

$$\boxed{N' > k^2 \frac{4(1-t')^2 - (1+x' - 2t')^2}{(x' - t')^2}} \quad (22)$$

for $x' \neq t'$. Using $x' = (x+1)/2$ and $t' = (t+1)/2$,

$$\boxed{N' > k^2 \frac{4(1-t)^2 - (1+x - 2t)^2}{(x - t)^2}} \quad (23)$$

for $x \neq t$. Since $y' = 1 - 2p'$, this also requires $y' \in [-1, 1]$, which is equivalent to

$$x \geq 4t - 3. \quad (24)$$

Comparison of Problems 2 and 3

From Solutions 2 and 3,

$$N > k^2 \frac{4 - (x-t)^2}{(x-t)^2}, \quad N' > k^2 \frac{4(1-t)^2 - (1+x-2t)^2}{(x-t)^2}. \quad (25)$$

Therefore

$$\frac{N}{N'} = \frac{4 - (x-t)^2}{4(1-t)^2 - (1+x-2t)^2}. \quad (26)$$

Hence there is no universal winner:

$$\frac{N}{N'} > 1 \iff 1 + 2x + 4t - 2xt - t^2 > 0, \quad (27)$$

that is,

$$N' < N \iff 1 + 2x + 4t - 2xt - t^2 > 0, \quad (28)$$

while $N' > N$ when the inequality is reversed.

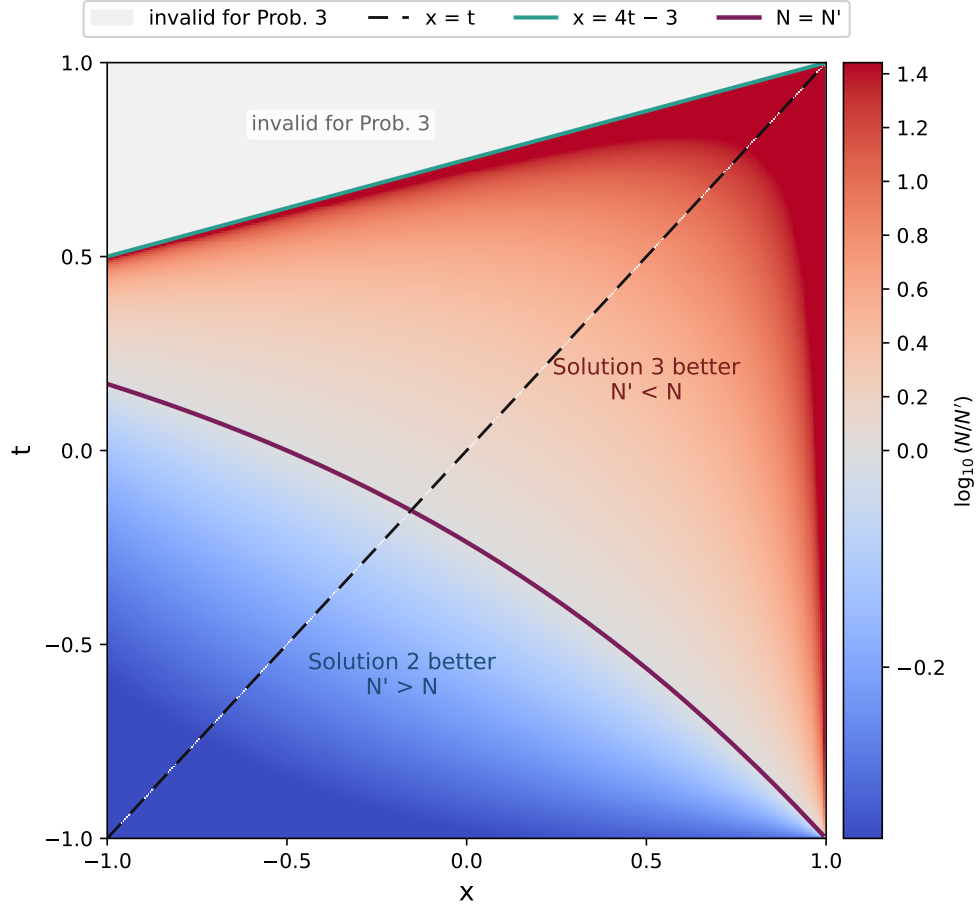


Figure 1: Comparison of the shot requirements for Solutions 2 and 3 over the (x, t) plane. The color scale shows $\log_{10}(N/N')$, so positive values indicate that Solution 3 requires fewer shots, while negative values indicate that Solution 2 requires fewer shots. The magenta contour marks $N = N'$. The dashed black line is $x = t$, where Solution 2 becomes singular, and the green line is $x = 4t - 3$, which bounds the valid domain of Solution 3. The gray region above the green line is invalid for Solution 3.